Lesson 8 Docker Repository

A Docker repository is a centralized location for storing images. Docker Hub (https://hub.docker.com/)" is the largest public repository, which stores tremendous images for users to download. Public repositories in China include Ali Cloud and Netease Cloud.

1. Publish Image to Docker Hub

① Address: https://hub.docker.com/. Register an account first.

② Ensure that you can successfully log in to your account.

③ Use the "tag" command to modify the image name.

The standard for publishing an image to "docker hub" is: "docker push" register username/image name

In this case, the registered username is "tao082", and the image name should be modified first.

1) Press "Ctrl+Alt+T" to open the command line terminal, and enter "docker images" to check the images on the master.

```
pi@raspberrypi:~ $ docker images
REPOSITORY     TAG       IMAGE ID       CREATED          SIZE
debian         1.1       30555f5a4f74   28 minutes ago   139MB
test/debian    v1        7f96d767c7be   18 hours ago     139MB
mysql          latest    b068ea59c677   4 days ago       638MB
debian         latest    60e0ade36b1a   12 days ago      139MB
ros            melodic   3365d517511a   6 weeks ago      1.2GB
hello-world    latest    ee301c921b8a   8 months ago     9.14kB
```

2) Enter "docker tag 30555f5a4f74 tao082/debian:1.1" in the terminal to modify the image names.

```
pi@raspberrypi:~ $ docker tag 30555f5a4f74 tao082/debian:1.1
```

3) Enter "docker images" in the terminal to see the modified image names.

```
pi@raspberrypi:~ $ docker images
REPOSITORY          TAG         IMAGE ID        CREATED            SIZE
debian              1.1         30555f5a4f74    35 minutes ago     139MB
tao082/debian       1.1         30555f5a4f74    35 minutes ago     139MB
```

① Log in to "docker hub" to publish the images:

4) Enter "docker login -u tao082" in the terminal and press "Enter". Input the password to log in.

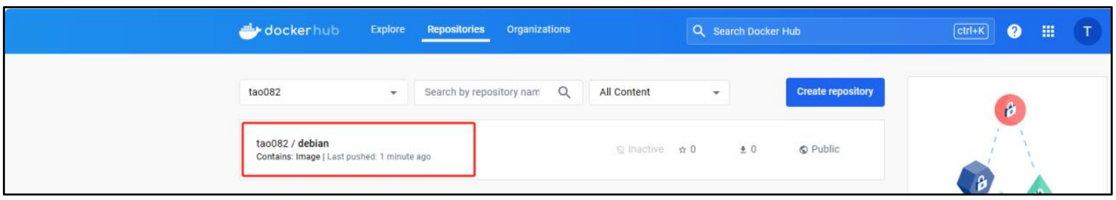
```
pi@raspberrypi:~ $ docker login -u tao082
Password:
WARNING! Your password will be stored unencrypted in /home/pi/.docker/config.jso
n.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
```

5) Enter "docker push tao082/debian:1.1" in the terminal and press "Enter" to publish the images.

```
pi@raspberrypi:~ $ docker push tao082/debian:1.1
The push refers to repository [docker.io/tao082/debian]
72c13eb508ab: Layer already exists
688a80a87f8f: Layer already exists
39ce3f073d94: Layer already exists
bf7f4fd8fe90: Layer already exists
77ad615d04d4: Layer already exists
1.1: digest: sha256:509237074c04151d36b3e082a709edf9a57cf54924f24c4913b5ecc1a350
6271 size: 1358
```

② Access the "docker hub", and you can view that the images have been successfully published.



2. Publish Image to Create “tar” Zip

The methods below are performed in the host, not in the Docker.

Method 1: “docker export” and “docker import” (export zip from containers)

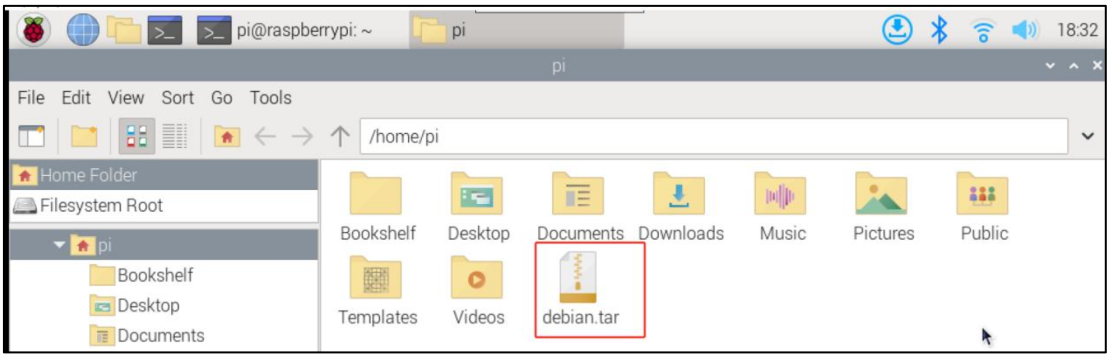
1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker ps” to display the latest containers in the master.

```
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE    COMMAND       CREATED         STATUS         PORTS    NAMES
efe969b704a7   debian   "/bin/bash"   8 seconds ago   Up 5 seconds            fervent_williams
cd1df225f0bf   debian   "/bin/bash"   8 minutes ago   Up 7 minutes            nostalgic_black
```

2) Enter “docker export efe969b704a7 >debian.tar” in the terminal to export the container “efe969b704a7” as “debian.tar”.

```
pi@raspberrypi:~ $ docker export efe969b704a7 >debian.tar
```

You can see the exported containers under the current path.



3) Enter “cat /home/pi/debian.tar | docker import - test/debian:v1” in the terminal. Import the path where the container is into “/home/pi/debian.tar”, name the image “test/debian” and the tag “v1”.

```
pi@raspberrypi:~ $ cat /home/pi/debian.tar | docker import - test/debian:v1
sha256:7f96d767c7be0fc55a3ab368058590df2245b1bae477cdf1a5fc39f2d46a72b0
```

4) Enter “docker images” in the terminal and press “Enter” to view the container information in the master.

```
pi@raspberrypi:~ $ docker images
REPOSITORY     TAG       IMAGE ID       CREATED          SIZE
test/debian    v1        7f96d767c7be   5 minutes ago    139MB
debian         latest    60e0ade36b1a   11 days ago      139MB
hello-world    latest    ee301c921b8a   8 months ago     9.14kB
```

Method 2: “docker save” and “docker load” (export zip from images)

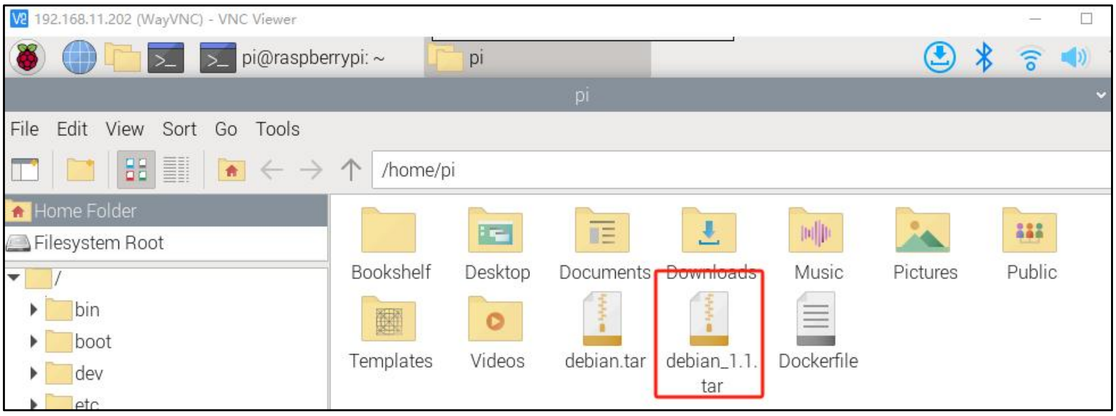
1) Press “Ctrl+Alt+T” to open the command line terminal, and enter “docker images” to view the latest images in the master.

```
pi@raspberrypi:~ $ docker images
REPOSITORY     TAG       IMAGE ID       CREATED          SIZE
test/debian    v1        7f96d767c7be   5 minutes ago    139MB
debian         latest    60e0ade36b1a   11 days ago      139MB
hello-world    latest    ee301c921b8a   8 months ago     9.14kB
```

2) Enter “docker save -o debian_1.1.tar debian:latest” in the terminal. Press “Enter” to create the images into a zip.

```
pi@raspberrypi:~ $ docker save -o debian_1.1.tar debian:latest
```

You can see the exported containers under the current path.



3) Enter “docker load -i debian_1.1.tar” in the terminal. After the zip is created, use “docker load” to import it into the image library.

```
pi@raspberrypi:~ $ docker load -i debian_1.1.tar
Loaded image: debian:latest
```

Differences between these two methods:

“docker export” is generated from the container, and “docker save” is created in the image. “docker export” saves smaller packages than “docker save”. The reason is that the latter saves the entire layered file system, while the former only exports one layer of the file system. Both “docker import” and “docker load” can be used to generate images. However, the former allows for renaming the images and specifying new version numbers, the latter cannot offer this option.